

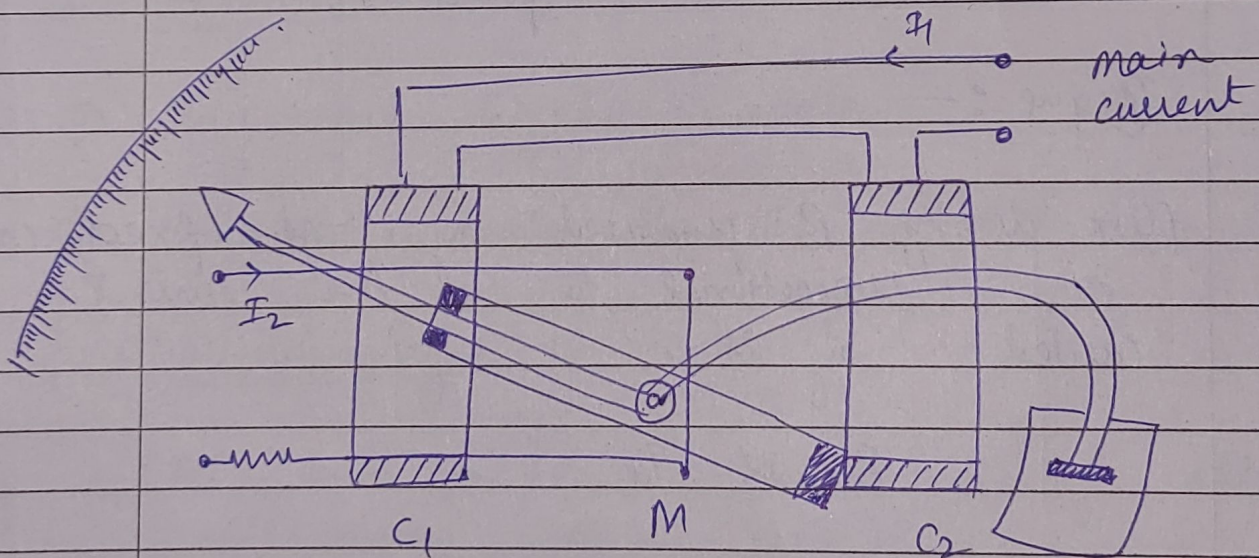
Unit-3

Dynamometer type Wattmeter instruments

- ↳ In this, no permanent magnet is used.
- ↳ The operating field is produced by a fixed coil.
- ↳ The moving coil is placed in the electromagnetic field of the fixed coil.
- ↳ The reaction between the fields produces the rotating force on the moving coil.

Construction :-

Consider,



- ↳ Fixed coil is made up of two sections C_1 and C_2 kept parallel to each other.
- ↳ The moving coil M is placed in the electromagnetic field of the fixed coil.
- ↳ The moving coil is attached to a spindle which carries a pointer.

- Springs are used for producing controlling torque. provided
- Damping is controlled by air friction, with the help of a light Al piston.

Operations :-

- The fixed coil carries the main current I_1 and the moving coil carries current I_2 which is proportional to the ~~current carried by it~~.
Supply voltage V .

Hence, the pointer depends on V and I_1 which means circuit power.

Torque :-

flux density B produced by the fixed coil is proportional to the current carried by it.

$$B \propto I_1$$

Also, $I_2 \propto V$

Therefore, $B I_2 \propto V I_1 \propto T_d$

$T_d \propto \text{power}$

→ DC

$$\boxed{T_d \propto VI \cos \phi \propto \text{power}} \rightarrow \text{Ac. chts}$$

For spring control,

$$T_c \propto \theta$$

At final deflected position,

$$T_d = T_c$$

$$\boxed{\theta \propto \text{power}}$$

→ Scales of dynamometer wattmeter are uniform.

Advantages :-

- ① They are highly accurate
- ② Equally accurate on dc as well as Ac.

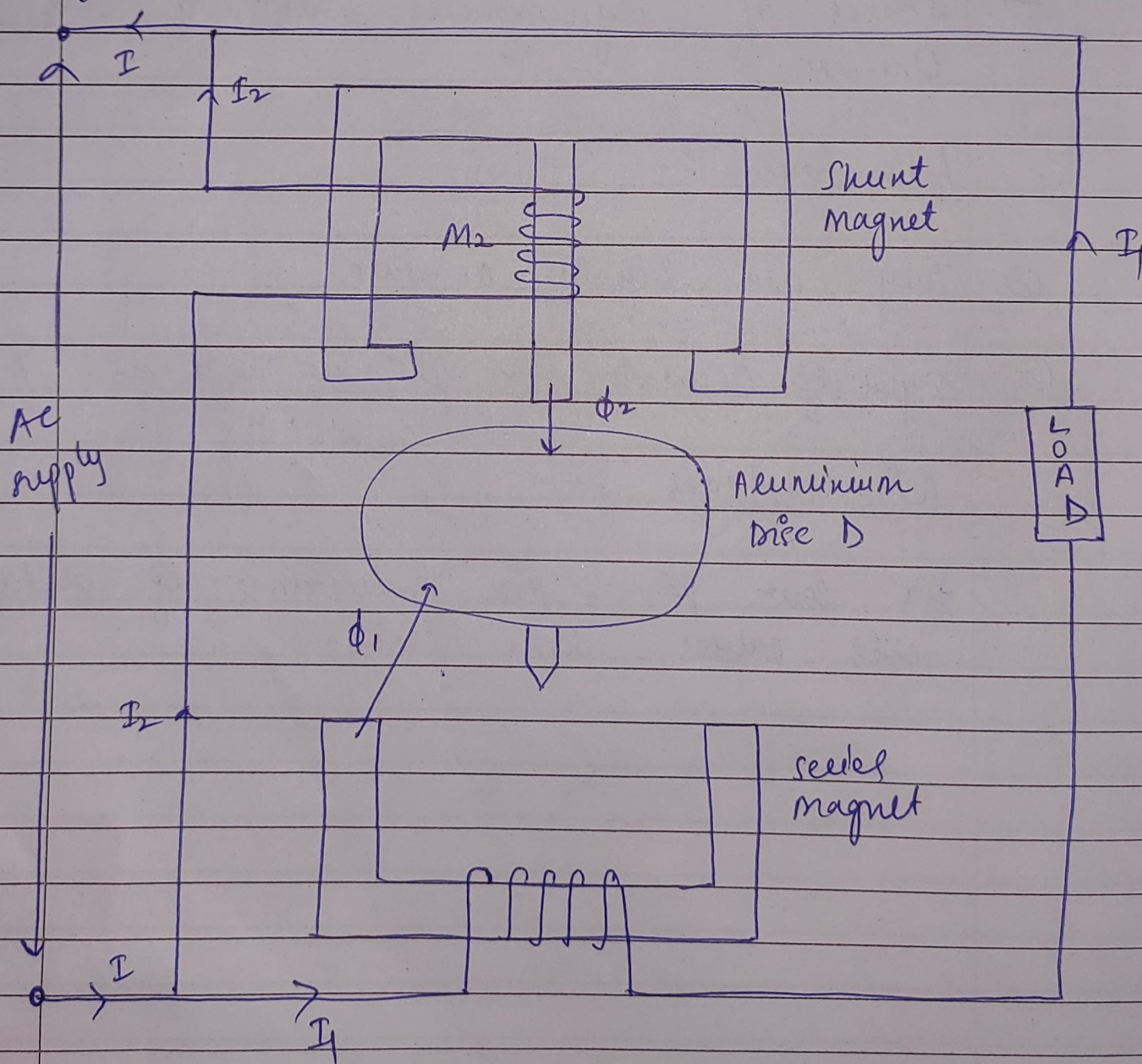
Disadvantages :-

At low pf, the inductance of voltage coil causes some error.

Induction type Wattmeters

- This wattmeter has two ^{AC} magnets, one excited by the line current and the other by the current proportional to the voltage.
- The two alternating fluxes and the eddy currents produced react with each other.
- And, this reaction is the basic principle of this wattmeter.

Construction :-



- Has two laminated electromagnets M_1 and M_2 .
- M_1 is connected in series with the ckt and carries current I_1 . Also called series magnet.
- M_2 is connected in parallel and carries current I_2 which is proportional to supply voltage V . Also called shunt magnet.

⇒ Since, a thin aluminium disc D cuts the fluxes of both magnets and hence, eddy currents are induced in it. The deflecting torque is produced due to the interaction of these eddy-currents with the two alternating fluxes ϕ_1 and ϕ_2 .

- A pointer attached to the spindle moves over the calibrated scale.

Deflecting torque :-

D. torque depends on two currents I_1 and I_2 -

$$T_d \propto I_1 I_2$$

As, $I_2 \propto V$

$$\boxed{T_d \propto V I_1}$$

$$\boxed{T_d \propto V I_1 \cos \phi \propto \text{Power}}$$

For spring controlled, $T_c \propto \theta$

~~For~~

final deflected position,

$$T_c = T_d$$

$$\theta \propto \text{power}$$

Scales of such wattmeters are uniform ;
equally spaced.

Advantages :-

- ① Have fairly long scales (over 300°)
- ② free from effects of stray fields.
- ③ Have good damping.
- ④ Almost free from freq. error.

Disadvantages :-

Only disadvantage $\rightarrow$ temp. error. due to
resistance of eddy currents.

1- ϕ Power factor meter

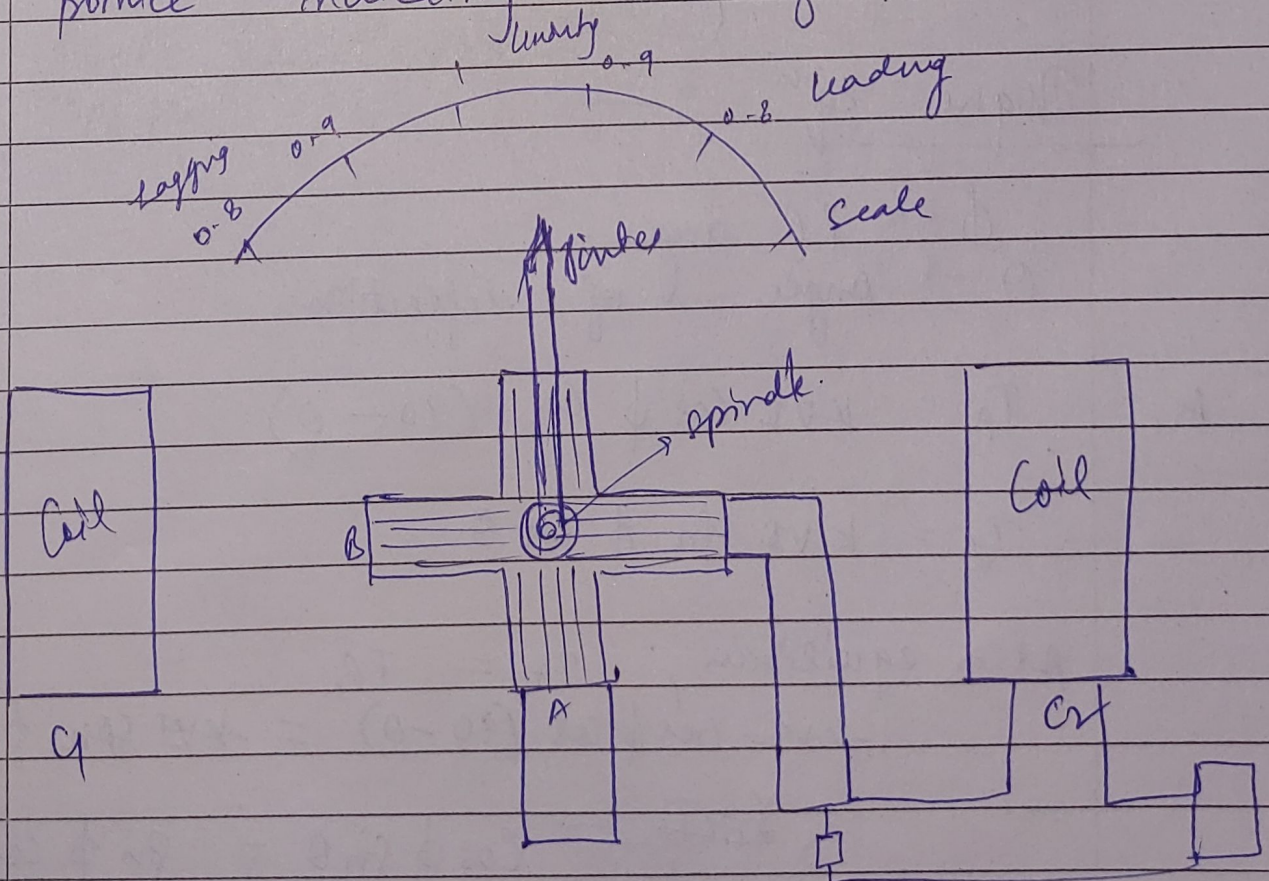
- Used to measure pf in TL.
- Cosine angle b/w V & I is called pf.

$$Pf = VI \cos \phi$$

- determines the connected load in transmission system.

Construction

- consists of fixed coil & moving coil.
- fixed coil divided into two equal halves.
- Two moving coil placed in middle remain at 90° from each other.
- A spindle is attached on the moving coil with a pointer.
- Both moving coils rotate together with pointer indicating value of pf on scale.



- fixed coil carry main current due to which MF produced around it.
- Moving coil also known as pressure coil or voltage coil.
- When 1- ϕ AC supply is given, current starts to flow. So, magnetic field produced in the fixed coil and flux induced in around moving coil.
- Due to interaction of both the fields, torque develops.
- The developing torque in A & B coil is opposite to each other.
- As, Inductance is connected with coil B, current in B lags 90° by voltage.
- When torque remain equal & opposite, pointer remains at equilibrium position, means unity Pf.

Torque Eqⁿ :-

$\phi \rightarrow$ If angle
 $\theta \rightarrow$ Angle of deflection

As,

$$T_A = KV I \cos \phi \cos (90 - \theta)$$

$$T_B = KV I \sin \phi \cos \theta$$

At equilibrium, $T_A = T_B$

$$KV I \cos \phi \cos (90 - \theta) = KV I \sin \phi \cos \theta$$

$$\cancel{\sin \theta} \cos \phi \sin \theta = \sin \phi \cos \theta$$

$$\sin \theta = \tan \phi \cos \theta$$

$$\tan \theta = \tan \phi$$

$$\boxed{\theta = \phi}$$

Advantages :-

- Less losses occurs as minimum iron used.
- Less error over a small range of freq.
- High torque & weight ratio.
- Less maintenance cost.
- Easy transportation.
- Used to pf in TL.

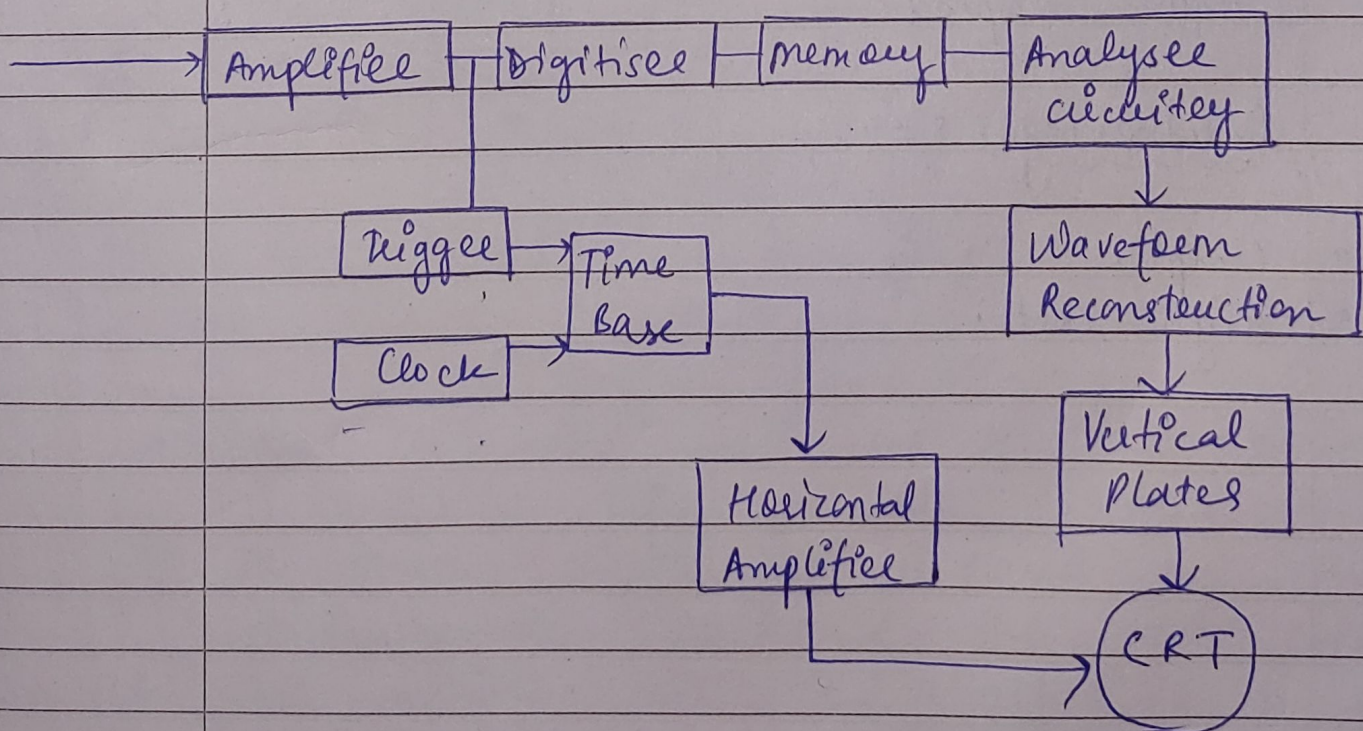
Disadvantages :-

- Small working forces.
- Scale can't extend till 360°
- Costly.

DSO

- The DSO is an instrument which gives storage of a digital waveform or digital copy of waveform.
- It allows us to store the signal in digital format.
- It allows us to do digital signal processing techniques over that signal.
- The traces on the DSO are bright, highly defined & displayed within seconds.
- It captures non-repetitive signals and displays it on the screen until the device gets reset.

Block Diagram :-



- firstly, analog signals enters into the amplifier, so that weak signals gets amplified.
- then, signals enter in the second stage which involves ~~receiving~~ digitisation of the analog signals into digital signals. It uses A/D converter for this purpose.
- _____ Working self.

→ Digitisation of input signals should follow sampling theorem.

Advantages :-

- Portable
- High bandwidth operation
- User interface is simple
- High speed

Disadvantages :-

- Complex
- High cost.